

Shaani Kamra Lyons

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EDUCATION

Queen's University Kingston, Ontario

Expected April 2028

Bachelor of Applied Science (B.A.Sc) — Electrical Engineering | GPA: 3.74

TECHNICAL SKILLS

Technical: KiCad, AutoCAD, LTSpice, Quartus II, SolidWorks, Arduino Systems, Soldering & Breadboard Testing, Adobe Lightroom

Programming: C, Python, VHDL, Assembly, Arduino Control Logic, Pseudocode

Transferable: Team Collaboration, Cross-Functional Communication, Time Management, Problem-Solving, Adaptability

PROJECTS

Custom STM32F411 Development Board

Aug 2026

- Designed a custom microcontroller development board in **KiCad**, pairing an **STM32F411** (Arm Cortex-M4) with an **ESP32** Wi-Fi/Bluetooth module and dual USB/battery power management for flexible bench testing.
- Completed a 2-layer **PCB** layout that passed all electrical and design rule checks on the first submission, then ordered a 5-unit prototype build for hands-on validation.
- Broke out **I2C**, **SPI**, and **GPIO** headers so future sensors can be added without redesigning the board, keeping the platform reusable for later projects.

16-Bit Calculator ALU — FPGA Implementation

Jul 2026

- Designed and implemented a 16-bit calculator **ALU** in **VHDL** on an **Altera DE0 Nano FPGA**, synthesized in **Quartus II**.
- Built and integrated an adder/subtractor, multiplier, and division state machine into one working **ALU**, using on-board **hardware testing** to check outputs against expected results.
- Debugged and resolved multiple design issues until all 4 arithmetic operations (add, subtract, multiply, divide) produced correct results directly on the FPGA.

Strain Measuring PCB — Queen's Baja SAE

Jan 2025 – Apr 2025

- Worked as part of a 5-person **team** on Queen's Baja SAE, designing a custom **PCB** in **KiCad** to measure mechanical strain via a Wheatstone bridge and strain gauge for the race vehicle.
- Coordinated closely with teammates across circuit design, calibration, and testing to reduce the board footprint by over 90% (7 cm × 7 cm → 3 cm × 2 cm) through iterative layout optimization.
- Built and calibrated a signal amplification circuit in **LTSpice**, confirming a 2x amplification factor through hands-on voltage testing, and helped the **team** deliver the prototype \$12 under the \$100 budget.

Automated Water Pre-Treatment System

Sep 2024 – Dec 2024

- Collaborated with a 5-person **team**, dividing work across mechanical, electrical, and software subsystems to build an **Arduino**-controlled prototype automating water treatment end-to-end.
- Designed and tested 3 dispensing assembly prototypes in **SolidWorks** alongside teammates, then programmed dosing control logic in **C**, jointly selecting the design that dispensed 0.7 g of alum on average within the 0.6–1.0 g target range.
- Worked with the **team** through iterative sensor calibration to achieve consistent turbidity readings of 3.8–4.4 V, meeting the project's water quality target.

WORK EXPERIENCE

Lakelands Irrigation Barrie, ON

May 2026 – Present

Irrigation Technician

- Installed, started up, and serviced 300+ residential and commercial irrigation systems, including electrical controllers and wiring.
- Diagnosed and resolved mechanical and electrical system issues across multiple job sites through hands-on troubleshooting and service calls.
- Worked closely with a crew **team** to design tailored irrigation layouts, coordinating on head, pipe, and fitting selection to meet unique pressure and layout requirements.

EngSoc Photography Team Kingston, ON

Oct 2025 – Present

Photographer

- Photographed engineering events, competitions, and conferences for EngSoc as part of a photography **team**, managing individual shoots independently through final delivery.
- Edited and processed images using **Adobe Lightroom** for publication across EngSoc's digital platforms, coordinating with the wider **team** to meet event deadlines.

AWARDS AND ACCOLADES

- Dean's Scholar at Queen's University
- Sir Isaac Newton Exam Distinction — University of Waterloo
- Ontario Scholar

INTERESTS

Skiing, Photography, Video Games, Hockey